MATH 201 Your name: ____________________

Learning target AD5 and AD4, version 1

AD5: Draw annotated number lines and apply Derivative Tests to classify the critical points as local extrema.
AD4: Use the Extreme Value Theorem to find the global maximum and minimum values of a continuous function on a closed interval.

(AD5)

Consider the function $f(x) = 2x^3 + 24x^2 - 54x + 7$.

1. Find the critical points using algebra, and draw a sign chart (aka labeled number line).
   Use the sign chart to decide what kind of extremum happens at each critical number.

(AD4)

2. Restricting the domain of this function to the interval $[-8, 2]$, what are the absolute maximum and absolute minimum values of this function, and where do they occur?

3. What if we change the domain to $[-10, 2]$?

MATH 201 Your name: ______________________

Learning target AD5 and AD4, version 2

AD5: Draw annotated number lines and apply Derivative Tests to classify the critical points as local extrema.
AD4: Use the Extreme Value Theorem to find the global maximum and minimum values of a continuous function on a closed interval.

(AD5)

Consider the function $g(x) = -2x^3 + 18x^2 + 42x - 100$.

1. Find the critical numbers using algebra, and draw a sign chart (aka labeled number line). Use the sign chart to decide what kind of extremum happens at each critical number.

(AD4)

2. Restricting the domain of this function to the interval $[-4, 4]$, what are the absolute maximum and absolute minimum values of this function, and where do they occur?

3. What if we change the domain to $[-4, 10]$?